# Does the Defense Still Hold? Three Boundaries for Reusing Side-Channel Defense Evidence in LLM and Agent Deployments

***Abstract*—Side-channel defenses for LLM serving and agents are evaluated in one setting and then reused in another: a different receiver, a different execution path, a differently trained attacker. We argue that three boundaries decide whether such reuse is sound: the *receiving view* (what the observer sees after the last emitter), the *affected state* (whether the defense only discards outputs or executes actions that change state), and the *attacker and workload* (whether the attacker was fitted on defended outputs, and on which population). Observer specification and adaptive attacks are established principles in website fingerprinting and adversarial-ML evaluation; the affected-state boundary appears once a defense issues cover actions through an agent's tools, and those traditions have no counterpart for it.**

**We turn the boundaries into a six-field reporting card whose output, for each claim, is a supported comparison, an incompatible one, or a named missing measurement, and we fill it completely for a published cache defense. Two case studies show that the attacker boundary and the affected-state boundary can each reverse a conclusion. In a synthetic TLS replay, response batching drives a clean-trained attacker to chance balanced accuracy (0.500, AUROC 0.678), yet an attacker trained on batched outputs reaches 0.732 (AUROC 0.773), close to the undefended 0.744 (0.800). In a retail-agent sandbox, read cover produces no detectable change in task success ($+0.7$ points, 95% interval $[-4.9, 6.1]$) while side-effecting write cover lowers it by 46.9 points, although both discard the cover response. The card is a proposal: we have not measured whether it improves analysts' decisions, and we specify the study that would.**

## 1. Introduction

Encryption hides what an LLM says, but not how it says it. Prefix-cache hits shorten time to first token [1], [2], streamed token lengths survive TLS [3], [4], and the network traffic of a local research agent leaks its user's prompt and traits [5]. Defenses follow: selective cache sharing [6], padding and batching of streamed responses [3], [4], and cover requests that blur an agent's real actions [5].

These defenses are evaluated once and reused often. A provider that adopts a published defense changes the model, the serving stack or the agent's tools; a later paper cites the defense's numbers as a baseline. Each reuse silently assumes that the original measurement still describes the new deployment. Often it does not. WRA protects a local research agent by issuing decoy *read* requests and discarding their replies [5]; evidence that this costs little task utility says nothing about cover that *writes*. Early Bird reports an overhead of about 6 ms, but that figure belongs to its semantic-cache prototype, not to the prefix rule it evaluates for protection [1, V-A–B]. The defense name stays the same; the evidence object changes.

Three reuse boundaries. We argue that three boundaries decide whether a defense result can be carried into a new deployment (Figure 1):

- **Receiving view.** A result protects one observation. If a later component re-emits the protected content—a content-aware frontend, a tool provider, a logging path—the observer sees a new trace and the result no longer applies.
- **Affected state.** Discarding a cover reply and executing a cover action are different interventions. Once cover actions run through an agent's tools, they can consume quotas, change records and alter what the real task can do next.
- **Attacker and workload.** Protection against a classifier trained on undefended outputs does not imply protection against one trained on defended outputs, and a calibration fitted on one model or population is not automatically valid on another.

What is new and what is inherited. Observer specification and adaptive attacks are long-standing principles in website fingerprinting (WF) and adversarial-ML evaluation [7], [8], [9], [10]; we do not claim them. Two things are specific to LLM and agent deployments. First, the affected-state boundary has no counterpart in those traditions: WF dummy packets and adversarial-example defenses do not change the state of the environment, whereas agent cover actions can. Second, LLM systems routinely insert content-aware components between the protected link and the observer, so the receiving view must be re-established after every emitter. Our contribution is to make these boundaries explicit, operational and measurable for this setting (Section 2).

Contributions.

- **A reuse framework** (Section 3) that organizes defense evidence by receiving view, affected state, and attacker and workload, illustrated with published LLM and agent defenses.
- **A reporting card and decision rule** (Section 4). Six fields link protection, function and cost to the observation, intervention and attacker they were measured

under. For each claim the card yields *supported*, *incompatible* or *unresolved*, and an unresolved outcome names the missing measurement. We fill the card completely for CachePrune [6] and apply it to six defense papers.
- **Two case studies** (Section 5), each isolating one boundary. *Case A* (attacker fitting): in a synthetic TLS replay, batching appears to defeat a clean-trained attacker, but an attacker trained on batched outputs recovers almost all undefended leakage. *Case B* (affected state): in a retail-agent sandbox, read cover produces no detectable change in task success, whereas side-effecting write cover lowers it by 46.9 percentage points.

Scope. This is a methodological position paper. It is not a systematic literature review: the source examples were selected to illustrate each boundary and support no frequency claims. We have not measured whether the card improves analysts' decisions or how reliably different analysts fill it; Section 6.1 specifies the study that would test both. All other limitations are collected in Section 6.2.

## 2. Setting and Positioning

### 2.1. What a defense result is

An *observer* sees an auxiliary signal—timing, lengths, cache hits, tool traffic—and tries to infer a *hidden property*, such as a prompt topic, a cached prefix or a user trait. A *defense* transforms the signal or adds cover. A published defense result reports up to three outcomes: *protection* (an attack metric under the defense), *function* (output quality or task success) and *cost* (latency, bytes, compute). Each outcome is measured under particular conditions: who receives which signal, where and how the defense executes, how the attacker was trained and tested, and on which workload.

*Reuse* means citing such a result to justify a defense in a deployment that differs in at least one of those conditions. Reuse is sound when the conditions that the claim depends on are unchanged, or when the change has been measured. The rest of this paper is about identifying which conditions a given claim depends on. We use two attack metrics throughout: balanced accuracy (BA), the mean of per-class recall at one decision threshold (chance is 0.5 for two classes), and the area under the ROC curve (AUROC), which measures how well scores rank positives above negatives across all thresholds.

### 2.2. Relation to established evaluation guidance

Table 1 places the three boundaries against existing guidance. The comparison is analytical: we state what each tradition addresses, not an empirical finding that its users miss these issues.

WF research has long required explicit observer models, adaptive attackers trained on defended traces, protection/overhead trade-offs and tests under environment shift [11], [7], [8], [12]. Adversarial-ML guidance requires adaptive attacks tailored to the defense and a clearly specified evaluation interface [9], [10], and benchmarks fix comparison protocols [13]. Formal approaches—differential privacy, quantitative information flow, constant-time programming and oblivious RAM—state their observation and leakage models precisely [14], [15], [16], [17]. Our attacker-and-workload boundary is the adaptive-attack principle applied to LLM side channels; Case A quantifies it, but it is not new.

The other two boundaries change character in LLM and agent systems. In WF, the observer's position is usually fixed by the threat model (for example, between client and guard). LLM pipelines instead contain content-aware components—frontends that regroup decoded text, tool providers, output formatters—that consume protected content and emit a fresh signal. A defense on the upstream link then says nothing about the final trace, so the receiving view must be bound after the last emitter rather than assumed once. The affected-state boundary has no WF or adversarial-ML analogue: dummy packets and perturbed inputs do not change the state of a remote system, whereas an agent's cover action can place an order, consume a modification quota or change a ledger. Function evidence for cover that is discarded does not transfer to cover that executes.

What the card adds, then, is not a new evaluation principle but a per-claim join: it requires that protection, function and cost be tied to the same receiving view, executed intervention and attacker fit before they are reused together.

## 3. Three Reuse Boundaries

This section states each boundary, shows it in published LLM and agent defenses, and ends with the reporting rule it implies. The source examples were chosen because they exhibit the boundary clearly; Appendix C describes how they were collected.

### 3.1. Receiving view

A protection result covers the signal that was measured, at the point where it was measured. It does not cover a signal produced later from the same content.

NetEcho makes this concrete [4]. Packet-size shaping on the upstream API link changes the observed sizes, but a plaintext-aware frontend then regroups the decoded tokens and emits a fresh size trace, reopening the channel that the upstream shaping closed. The same issue arises within a single interface. SICC shows that once message content is authorized for release, its length and structure can still support inference, and isolates that form-only channel [18, Table 3]. DP-MemView protects attributes at its compliant semantic interface; a separate diagnostic that exposes an internal score reaches an attribute AUC of 0.976 [19, RQ2]. That number is not an attack on the compliant interface—it is a result for a different released view. A reuse decision that cites DP-MemView's privacy result must name which of the two views the deployment exposes.

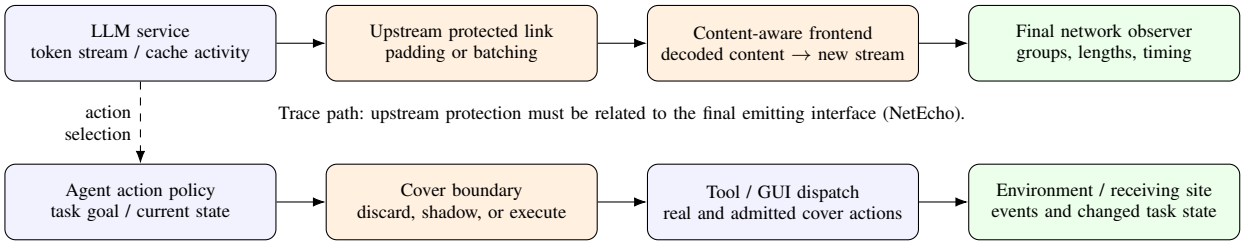



Figure 1. Where reuse breaks. Top: protection applied on an upstream link does not cover a trace that a later content-aware emitter produces. Bottom: once cover actions are admitted to tool or GUI execution, they can change the state that the real task depends on. A defense result must be bound to the trace that is actually observed and the action that is actually executed.

TABLE 1. THE THREE REUSE BOUNDARIES RELATIVE TO ESTABLISHED EVALUATION GUIDANCE.

| Boundary | Established in | What changes in LLM/agent deployments | Evidence in this paper |
|---|---|---|---|
| Receiving view | WF observer models [7], [8]; formal observation models [16], [15] | Content-aware emitters re-encode protected content after the protected link; authorized semantic content can still leak through form, length or exposed scores | NetEcho, SICC, DP-MemView (§3.1) |
| Affected state | No direct counterpart: WF cover and adversarial perturbations do not change remote state | Cover actions executed through tools can change records, quotas and the real task's options | Case B (§5.2); WRA, AHB (§3.2) |
| Attacker and workload | Adaptive attacks [10], [9]; WF training on defended traces [7], [11] | Defenses are calibrated per model or per workload; a source calibration is not a target calibration | Case A (§5.1); Time Will Tell, Whisper Leak (§3.3) |

**Rule 1 (Receiving view).** Bind each protection result to the final emitter and the exact fields it releases. If the deployment adds an emitter or releases another field, measure inference on that trace.

## 3.2. Affected state

Cover is the main defense against agent traffic analysis, and it comes in two kinds. WRA issues decoy research requests and returns only the real output to the user; the decoys' replies are discarded [5]. AHB humanizes mobile GUI agents by executing additional GUI events [20]. A reply that is discarded cannot affect the task. An event that executes can.

This difference matters because "the response was discarded" is a statement about content, while task success is a statement about state. A write whose response is ignored still commits; a sandbox, dry run or rollback limits that effect only within its own contract [21], [22], [23], and some changes, such as sequence advances, survive rollback [24]. Other agent defenses preserve narrower objects: UI-trace timestamp edits preserve action order [25], and Dangol's protected captures preserve fixed message bodies [26]. Each supports its stated preservation claim, and none establishes task completion under added execution. Case B (§5.2) measures how large the difference between discarded and executed cover can be.

**Rule 2 (Affected state).** State whether cover is discarded, isolated or executed. A content-preservation witness supports only the preserved content; executed cover needs a task-outcome measurement that includes its state effects, quotas and failures.

## 3.3. Attacker and workload

A protection number depends on how the attacker was trained. A classifier fitted to undefended outputs and then tested on defended outputs measures distribution shift as much as protection; the relevant attacker is fitted on defended outputs [10], [7]. Case A (§5.1) shows the gap in an LLM streaming setting.

Workload changes raise a related question: what exactly was held out? Table 2 separates three evidence objects that are often conflated. LLM-Fin, ArchSnoop and LeakyLMs show that *attacks* generalize to unseen model variants or architectures [27], [28], [29]. Time Will Tell instead evaluates one *defended condition* on several models: a 60-word-explanation instruction changes mean attack success from 88.2% to 61.9% on GPT-4o, but from 79.3% to 80.3% on Gemma2-9b-IT [30, VII-B]. Whisper Leak reports mitigations per provider and model [3, 3.2,6]. Such results are local to each model; they are not evidence that a defense calibrated on one model protects another. That third object, *defense transfer*, requires holding a source calibration fixed and comparing it with one fitted locally on the target.

**Rule 3 (Attacker and workload).** Report whether the attacker was fitted on defended outputs and which unit was held out. Reuse an attack-generalization result to choose the attacker, a defended-condition result for its own model, and a defense-transfer result only along the tested direction.

## 3.4. Function and cost follow the configuration

The three boundaries also govern function and cost. A cost or quality number is reusable only for the configu-

TABLE 2. THREE EVIDENCE OBJECTS FOR INTERPRETING HOLDOUTS. THE SAME FAMILY MAY SUPPLY SEVERAL OBJECTS IN SEPARATE EXPERIMENTS.

| Object | Required link | Permitted reuse |
|---|---|---|
| Attack generalization | Training/profiling source → held-out inputs or systems; attack outcome. | An attacker or recognition method covers the stated target domain. |
| Defended condition | Named mechanism, local fitting, defended test and outcome. | Retain that operating point; multi-model testing alone is not held-out transfer. |
| Defense transfer | Source-selected policy/calibration held fixed → target; local-calibration comparator and attacker fit specified. | Transport the evaluated policy only over the tested direction and scope. |

ration that produced it. Early Bird's approximately 6 ms de-identification overhead was measured on a semantic-cache prototype with 1,000 Presidio sentences, while its token-recovery results (91.5% at $K{=}1$, 49.0% at $K{=}4$; v5 Table VI) concern a minimum-shared-prefix rule [1, V-A–B]. The overhead cannot price the prefix rule, and we did not locate a serving-cost or task-utility measurement for it. SpliceDefense reports a sub-1% prefill cost for quantized chunk padding, a different component from the boundary-fusion mechanism whose timing residuals it reports [31, VII-C, Table II]. Formal and hardware defenses keep their own boundaries: OTRO's DRAM-trace indistinguishability holds under its ORAM and pseudorandom-function assumptions and excludes computation timing [32], and Palermo's near-zero mutual-information statistic concerns its stated stash/tree and timing variables under an external-memory observer, not prompt recovery [33]. In each case the source supports its own claim; the error would lie in the join.

# 4. The Reporting Card

The three rules become usable only if a reader can apply them to a specific paper and a specific deployment. The reporting card (Table 3) records, for one source configuration, the evidence for each boundary together with the function and cost measured under it.

## 4.1. From fields to a decision

A reuse decision starts from a *claim*, not from a paper: "this defense protects my deployment's TTFT channel", "its quality loss is negligible on my model", "its overhead is 6 ms". Each claim depends on a subset of the fields. A protection claim needs Receiver, Intervention, Attacker and Extension; a cost claim needs Intervention, Cost and Extension; a function claim for executed cover needs Intervention and Function at the task level. The decision rule is:

- **Supported** if every required field is **B** and the deployment matches it.
- **Incompatible** if a required field shows a demonstrated mismatch, for example a cost measured with the protection disabled.
- **Unresolved** otherwise. The output names the fields that are **P**, **U** or **?**, and therefore the measurement still needed.

TABLE 3. THE REPORTING CARD. FILL ONE CARD PER SOURCE CONFIGURATION AND PROPOSED DEPLOYMENT. EVERY EVIDENCE CELL CARRIES A SOURCE VERSION AND SECTION/TABLE LOCATOR AND ONE STATUS CODE.

| Field | What to enter |
|---|---|
| Receiver/target | Hidden property; the signal and fields the observer receives; observer permissions. |
| Intervention | Mechanism; where it executes (stored trace, wire, tool action); whether cover is discarded, isolated or executed. |
| Attacker/test | Data the attacker or calibration was fitted on; defended test population; held-out unit. |
| Function | Preservation witness, output quality or task outcome, at the configuration used for protection. |
| Cost | Resource, unit, baseline, denominator and included operations. |
| Extension | What the deployment changes (model, workload, hardware, network); which of these the source held out. |

**Status codes. B** bound: every element the claim needs is stated for this configuration. **P** partial: some are stated, at least one is not. **U** not located in the inspected sections. **?** stated but not verified. **–** not applicable to the claim.

Figure 2 shows the procedure. Three outcomes may seem coarse, but the information lies in their combination. One paper rarely yields a single verdict; it yields different outcomes for different claims, and an unresolved outcome is actionable because it specifies what to measure.

## 4.2. A complete example

Table 4 fills every field for CachePrune [6], which excludes detector-marked sensitive spans from cross-user KV-cache sharing. The hypothetical deployment runs a different model and its tenants can time responses remotely. The card separates three conclusions that a summary such as "CachePrune protects prompts at little cost" would merge. The recovery results support protection against binary hit/miss leakage, because the source's abstract oracle is at least as strong as a remote timer for that signal. The fixed-chunk comparison in §5.7 cannot price protection, because it disables privacy filtering. Quality, latency and detector recall must be measured again on the target. Filling the card required CachePrune's §3.1, §5.1–5.7 and Tables 1–2.

## 4.3. Applying the card to defense papers

Table 5 applies the card to six defense papers. All five for which output quality applies report a functional score, and the sixth (CacheSolidarity) supports exact-prefix preservation, so the card's role is not to find missing measurements.

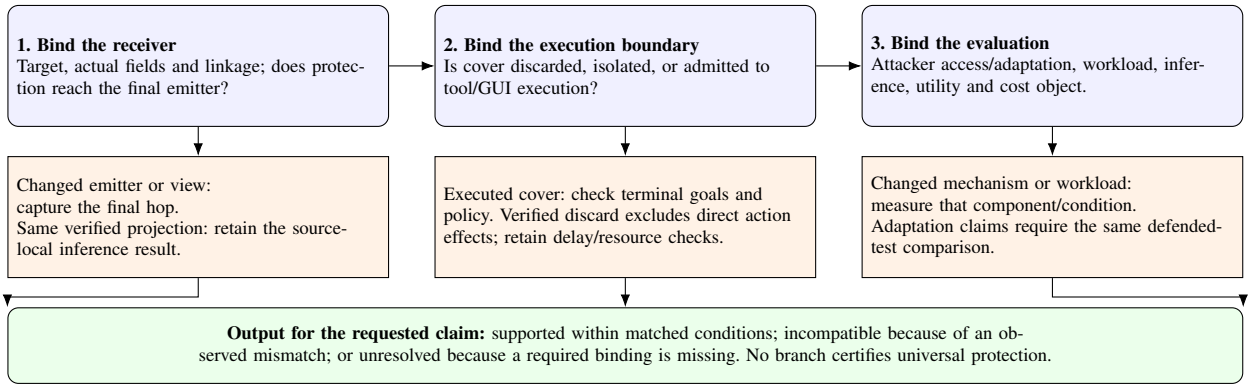



Figure 2. Decision procedure for reusing a defense result. Each step binds one boundary; the output is per claim.

TABLE 4. A COMPLETE CARD: REUSING CACHEPRUNE V1 [6] FOR A HYPOTHETICAL MULTI-TENANT SERVICE THAT RUNS A DIFFERENT MODEL AND WHOSE TENANTS MEASURE TIME TO FIRST TOKEN (TTFT) REMOTELY. ONE SOURCE YIELDS ALL THREE OUTCOMES, EACH TIED TO A SPECIFIC CLAIM.

| Field | Source evidence (locator) | Code | Proposed deployment | Outcome for this claim |
|---|---|---|---|---|
| Receiver/target | Abstract binary cache-reuse oracle; exact recovery of detector-marked sensitive spans on QASPER (§3.1, Table 1). | B | Remote tenants time TTFT over a WAN. | **Supported** for binary hit/miss leakage: a perfect oracle reveals at least what a noisy timer does. **Unresolved** for timing that reveals more than hit/miss, such as partial-hit length. |
| Intervention | Detector-marked spans are excluded from cross-user KV-cache sharing (§3.1). Detector recall on other data not reported. | P | Same mechanism, different user data. | **Unresolved**: protection is conditional on the detector; measure its recall on deployment data. |
| Attacker/test | Direct recovery 0%, contextual recovery 3.21% (Table 1). No learned attacker, so a clean-versus-defended fit is inapplicable; broader adaptive attacks not located. | B / – / U | Same two attacks; adaptive tenants possible. | **Supported** for the two evaluated attacks; **unresolved** for attacks that exploit unmarked context. |
| Function | QASPER F1 0.28 vs. 0.30 without sharing, Mistral-7B, default configuration (Table 2). Same configuration as Table 1, but paired runs not established. | P | Different model. | **Unresolved**: re-measure quality on the target model. |
| Cost | TTFT 273 vs. 797 ms without sharing, one H100 (Table 2). The fixed-chunk comparison (§5.7) disables privacy filtering. | P | Different model and hardware. | §5.7: **incompatible** as a cost of the protected configuration. Table 2: **unresolved** for the target hardware. |
| Extension | One model, one GPU type, one dataset; no model or hardware holdout located. | U | Model, hardware and network change. | Measure on the target. |

**Decision.** Retain the oracle-level recovery result as protection evidence for binary hit/miss leakage. Do not cite §5.7 as the cost of protection. Before deployment, measure detector recall on deployment data, quality and TTFT on the target model and hardware, and leakage from timing features beyond hit/miss. Inputs: CachePrune v1 §3.1, §5.1–5.7, Tables 1–2.

It shows which measurements may be joined. OCELOT binds protection, task success and an 18.3% wall-clock overhead at the same release boundary [34], so the three can be reused together. PlanTwin's re-identification diagnostic uses unrestricted fields, so it cannot be joined to planning quality under the privacy budget [35]. DP-MemView's AUC of 0.976 applies to the exposed-score view only (Rule 1). SICC and OCELOT evaluate adaptive attacks beyond the clean-versus-defended fitting contrast; recording them under Attacker/test keeps that evidence visible rather than reducing it to a single adaptive/non-adaptive flag.

## 4.4. Filling, disagreement and adoption

Procedure. (1) State the claim and the deployment change. (2) For each required field, quote the source with a locator, or write "not located". (3) Assign a status code. (4) Derive the outcome and, if it is unresolved, the missing measurement. Inspected sections are recorded, so a **U** means "not located in these sections", which another reader can check.

Disagreement. When two readers fill the same card, we propose that each cell take the weaker code (**B** $>$ **P** $>$ **U**) until one reader produces a locator that settles it; a disagreement that remains is reported in the card rather than resolved by authority. This rule is a proposal; its effect on agreement has not been measured (Section 6.1).

Who uses the card. Defense authors can include a filled card as an appendix, stating which receiver, execution mode and attacker fit each number belongs to. Reviewers can ask three questions—which emitter's trace was measured, does cover execute, and was the attacker fitted on

TABLE 5. THE CARD APPLIED TO SIX DEFENSE PAPERS. EACH ROW RECORDS WHICH RECEIVER THE PROTECTION RESULT COVERS, WHICH FUNCTION WAS MEASURED WITH IT, AND WHAT LIMITS THE REUSE OF ITS COST OR TRANSFER EVIDENCE. VERSIONS AND SECTION LOCATORS ARE IN THE SUPPLEMENT (POST-HOC CHECKS).

| Source | Receiver / protection evidence | Function measured | Cost / transfer boundary |
|---|---|---|---|
| CacheSolidarity [36] | Cross-user TTFT; 20-probe timing demonstration, not a scored recovery rate. | Exact-prefix preservation (logical). | TTFT, hit rate and component costs; model and load varied, calibration not held fixed. |
| PlanTwin [35] | Cloud-visible planning schema; re-identification measured with unrestricted fields, not under the privacy budget. | Planning quality. | Timing of heuristic default and small-model variant needs a configuration match. |
| SICC [18] | Form-only (length/structure) inference isolated; adaptive transcript attacks. | Protocol completion per receiver. | No compiler LLM calls; runtime cost partly measured. |
| DP-MemView [19] | Semantic transcript inference at the compliant interface; separate exposed-score diagnostic (AUC 0.976). | Response quality and personalization. | Runtime cost not bound; corpus transfer fixes scaffold and auditor. |
| OCELOT [34] | Semantic trajectory inference; held-out stress proxies and adaptive attacks. | Task success, same condition. | 18.3% wall-clock overhead at the same boundary. |
| AirGapAgent [37] | Contextual extraction by a third party, not passive metadata inference. | Appropriate-answer utility. | Runtime cost not bound; model and scenario variation, calibration not held fixed. |

defended outputs—that correspond to Rules 1–3. Benchmark designers can log the same fields with every result, so that later reuse does not depend on reconstructing them from prose. An editable template accompanies the artifact.

## 5. Two Case Studies

Each case isolates one boundary and asks whether ignoring it would change the conclusion a reader draws from a defense result. Both are demonstrations in a controlled setting, not estimates of a defense's effect in deployment; their limits are stated at the end of each case and in Section 6.2. The supplement identifies Case A as E4 and Case B as R44.

### 5.1. Case A: attacker fitting under response batching

Question. Streaming LLM APIs leak token lengths through TLS record sizes and timing [3], [4]. Batching several chunks per record is a natural defense. If it is evaluated against an attacker trained on undefended traffic, how far does the reported protection differ from protection against an attacker trained on batched traffic?

Setup. Three pinned local models (qwen3:8b, qwen3:0.6b, llama3.2:1b) answer five prompt templates for each of 100 synthetic topics under a 128-token budget. The hidden property is binary: 50 medical versus 50 nonmedical topics. Saved output chunks are replayed through timed TLS 1.3 loopback with byte-identical decoded content, and the attacker observes response record lengths and timing. Evaluation uses five folds that hold out ten topics of each class with all their templates. For each defended test fold, $D_0$ is fitted on clean (undefended) training traffic and $D_2$ on defended training traffic; both are tested on the same defended outputs. We report the default operating point, Qwen 8B with four chunks per batch, using one fixed logistic-regression attacker.

TABLE 6. CASE A AT QWEN 8B, FOUR-CHUNK BATCHING: ONE FIXED LOGISTIC-REGRESSION ATTACKER, 500 TOPIC-HELD-OUT OUTPUTS. BRACKETS: PAIRED TOPIC-STRATIFIED BOOTSTRAP 95% INTERVALS, HOLDING FITTED PREDICTORS FIXED. THE RAW ROW USES THE SAME LEARNER; ITS DIFFERENCES FROM THE DEFENDED-FIT ROW ($-0.012$ BA, $-0.027$ AUROC) HAVE NO INTERVAL.

| Condition | BA | AUROC |
|---|---|---|
| No defense (raw stream) | 0.744 | 0.800 |
| Batched; attacker fit on clean ($D_0$) | 0.500 | 0.678 |
| Batched; attacker fit on batched ($D_2$) | 0.732 | 0.773 |
| $D_2 - D_0$ | $+0.232$ [0.168, 0.286] | $+0.094$ [0.005, 0.189] |

Result. Table 6 shows the reversal. Evaluated with $D_0$, batching looks like a strong defense: BA falls from 0.744 to chance. Evaluated with $D_2$, it barely helps: BA is 0.732 and AUROC 0.773, against 0.744 and 0.800 without defense. The paired difference between the two attackers is 0.232 BA [0.168, 0.286] and 0.094 AUROC [0.005, 0.189]. The pattern holds under five further topic-level reshuffles ($D_0$ AUROC 0.676–0.751, $D_2$ 0.770–0.785) and on a separate 20-topic holdout (0.758 versus 0.807).

The $D_0$ AUROC of 0.678 shows that the clean-trained attacker still ranks outputs better than chance; its chance BA is a threshold effect. That is part of the lesson. A defense paper that reports BA for a clean-trained attacker would publish "chance", and a reader who reused that number would conclude that batching defeats the channel. Both the attacker fit and the metric must travel with the result.

Cost. Cost is bound to the same 500 replays. Batched response application bytes total 470,239 versus 1,234,823 for paired raw captures ($0.381\times$), because aggregation removes record framing. Median session completion changes by $+0.152$ ms (95th percentile $+2.381$ ms), while the 95th-percentile delay of original chunk endpoints is $+65.730$ ms. These figures exclude requests and handshakes and describe loopback replay, not online generation or a WAN.

What the case shows and does not. The principle that adaptive attackers must be fitted on defended data is well

TABLE 7. CASE B TASK SUCCESS. EACH ARM HAS 187 EPISODES FROM 98 TASKS. TASK MEANS WEIGHT TASKS EQUALLY AFTER AVERAGING THEIR REPEATS; INTERVALS ARE 95% TASK BOOTSTRAPS; $p$ VALUES ARE SIGN-FLIP TESTS, HOLM-ADJUSTED OVER THE TWO CONTRASTS.

| Arm | Succ. | Task mean (%) | Diff. (pp) | $p$ |
|---|---|---|---|---|
| No cover | 154 | 82.7 [75.5, 89.3] | – | – |
| Read cover | 156 | 83.3 [76.5, 89.6] | $+0.7$ $[-4.9, 6.1]$ | 0.859 |
| Write cover | 72 | 35.7 [26.7, 45.1] | $-46.9$ $[-56.8, -36.9]$ | $<0.001$ |

known [10], [7]. The case shows how much it matters for a natural LLM streaming defense: the clean-trained evaluation would overstate protection by nearly the whole undefended signal. The corpus is synthetic and small, the task is binary, and the fixed-logistic comparison was chosen after earlier results on the same data were visible, so it is exploratory. We also compared calibrations imported across models with locally fitted ones; most such comparisons were degenerate because the parameters were identical, and the rest were inconclusive (Appendix A).

### 5.2. Case B: read cover versus write cover

Question. WRA-style cover discards the replies of its decoy reads, so its function evidence is that the user's output is unchanged [5]. Would the same evidence license cover that issues a write whose response is likewise discarded? A content-preservation witness treats the two identically. Rule 2 says it should not.

Setup. We ran a three-arm study in a retail-agent sandbox. In each episode an LLM planner serves a simulated user; after the first eligible order lookup, the arms add nothing, one hidden read, or one hidden write that modifies the same item. The cover response is discarded in both cover arms, but the write's effect on the payment ledger and the order's modification eligibility persists. Each arm uses fresh planner and user interactions on a separate clone of the initial state, and the arm order is randomized. The planner is a Qwen model accessed through a service alias whose weights are not pinned. The cohort is 187 three-arm groups (561 episodes) covering 98 tasks: the blocks left incomplete when an earlier run with a different planner (GLM) was interrupted, a nonrandom convenience sample. We weight tasks equally after averaging each task's repeats.

Result. Read cover produces no detectable change: $+0.7$ points, with a 95% task-bootstrap interval of $[-4.9, 6.1]$, which does not exclude effects of about five points in either direction. Write cover lowers it by 46.9 points $[-56.8, -36.9]$, from 82.7% to 35.7% (Table 7). All 97 executed cover writes consumed the order's modification eligibility. In 28 of the 115 failed write-arm episodes, a later real action on the same order received the sandbox's native "not pending" error; the other failures have not been attributed to a cause. The 113 blocks that the interrupted GLM run did complete, over 80 tasks, show the same pattern with a different planner: $+3.1$ points for read cover $[-6.5, 12.7]$ and $-45.6$ for write cover $[-57.9, -32.7]$. These blocks are not a random sample either, and they are not paired with the Qwen arms.

What the case shows and does not. The direction of the effect follows from the construction: a write that consumes a one-time modification right will block a later modification. The case is therefore a counterexample, not a measurement of a defense's cost. Its point is the gap between two arms that the same preservation evidence would certify: no detectable loss for read cover and a loss of nearly half of all successes for write cover. It does not estimate the cost of WRA or of write cover in general, and without a side-effect-free write arm it cannot apportion the loss between state effects and the planner's reaction to an extra call. During recovery of the original run, concurrency changed from 10 to 2 and the transport path changed; ten interrupted requests have unknown outcomes (Appendix B).

## 6. Validation, Limitations and Ethics

### 6.1. What has and has not been validated

The paper establishes two things. The source examples show that each boundary occurs in published LLM and agent defenses, and the two cases show that ignoring the attacker or affected-state boundary can reverse a conclusion by a large margin. It does not establish that analysts who use the card make better reuse decisions than analysts who use existing guidance, or that different analysts fill the card consistently. All cards in this paper were filled retrospectively by the authors with model assistance and with the source outcomes visible.

A study that tests both questions would proceed as follows. Recruit analysts with security-evaluation experience and randomly assign each to the card or to established guidance [9], [8], with counterbalanced case order and a common time budget. Use reuse scenarios built from sources with known join errors, such as the Early Bird overhead, CachePrune §5.7 and DP-MemView's exposed score, mixed with scenarios where reuse is sound. Score decisions against a reference adjudicated in advance by assessors blind to condition, and report decision accuracy, inter-coder agreement (for example, Cohen's $\kappa$ per field), completion time and their uncertainty, with participant and case effects modeled. Pilot the rubric and register the protocol before collecting outcomes. Until such a study is run, the card should be read as a structured proposal grounded in existing principles.

### 6.2. Limitations

Sources. The source examples were selected purposively, starting from attack papers that met five inclusion conditions and extended by a small defense-first search (Appendix C). They illustrate the boundaries and support no claims about how often the field reports any field of the card. Source coding was neither preregistered nor independently double-coded. Statements that a measurement was "not located" refer to the inspected version and sections. Several

sources are preprints whose cited version differs from the latest one; Appendix C lists the version questions that affect interpretation.

Case A. The corpus is synthetic (100 topics, five templates), the task is binary, the transport is TLS loopback rather than a WAN, and the defense is applied to saved outputs rather than during generation. The fixed-logistic comparison and the additional splits were defined after earlier results on the same data were visible; their intervals hold fitted predictors fixed and omit model-selection uncertainty. A 500-session repeat was collected in a new runtime after an incomplete original run.

Case B. The planner is a mutable service alias, the 98 tasks are a nonrandom convenience sample, and only one state-changing write is tested in one domain. There is no side-effect-free write control, and the recovery of the original run changed concurrency and transport, leaving ten requests with unknown outcomes.

### 6.3. Ethics

The source analysis uses public research papers. The cases use synthetic topics, cloned benchmark databases, simulated users and local sockets; no production accounts were modified and no human subjects were involved. Describing which observation a defense leaves uncovered can help an adversary choose a surface, so the artifact omits raw traces, credentials and operational detail while retaining source locators, derived labels, protocols, code and aggregate results. Redistribution of third-party papers and traces is subject to their terms.

### 6.4. Use of generative AI and reproducibility

AI tools assisted research preparation, implementation, source reconciliation and prose. OpenAI GPT-6 Codex assisted analysis code, citation-inventory preparation and prose in an earlier revision; the complete earlier assistant history is not recoverable. Anthropic Claude assisted the present restructuring of the manuscript and an automated check of reference metadata. Model assistance extends to source coding: cards were drafted with model help, and the authors are responsible for every code and interpretation. Because no independent coder checked them, each evidence cell carries a locator so that readers can verify it directly.

On 4 November 2026 we checked every reference that carries an arXiv identifier against the arXiv API: all 80 matched in title and first author. Of 29 references with a DOI, 27 resolved through Crossref with matching titles; two DOIs cited only in the supplement did not resolve and are recorded in the artifact's citation checks. Case A uses pinned local model artifacts; Case B used a mutable alias, so a fresh run can reproduce the protocol but not the model. Archived inputs, saved predictions, configuration hashes and analysis code allow the reported statistics to be recomputed without new model calls.

## 7. Conclusion

A side-channel defense result describes one observation, one execution and one attacker. Reusing it in an LLM or agent deployment is sound only when those three things still hold, and the reporting card makes that check explicit for each claim: supported, incompatible, or unresolved with a named measurement. Two cases show what is at stake. Against a clean-trained attacker, response batching looks like it reduces leakage to chance; against an attacker trained on batched outputs, it leaves nearly all of it. Cover whose response is discarded can still cost almost half of an agent's task successes when the cover writes. Whether the card helps analysts avoid such errors is the question that the next study must answer.

## References


[1] L. Song, Z. Pang, W. Wang, Z. Wang, X. Wang, H. Chen, W. Song, Y. Jin, D. Meng, and R. Hou, "The early bird catches the leak: Unveiling timing side channels in LLM serving systems," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 11 431–11 446, 2025, cited version: arXiv:2409.20002v5. [Online]. Available: https://ieeexplore.ieee.org/document/11206380/

[2] G. Wu, Z. Zhang, Y. Zhang, W. Wang, J. Niu, Y. Wu, and Y. Zhang, "I know what you asked: Prompt leakage via KV-cache sharing in multi-tenant LLM serving," in *Network and Distributed System Security Symposium (NDSS)*, 2025. [Online]. Available: https://www.ndss-symposium.org/ndss-paper/i-know-what-you-asked-prompt-leakage-via-kv-cache-sharing-in-multi-tenant-llm-serving/

[3] G. McDonald and J. B. Or, "Whisper leak: a side-channel attack on large language models," *arXiv preprint arXiv:2511.03675*, 2025, cited version: arXiv:2511.03675v1. [Online]. Available: https://arxiv.org/abs/2511.03675v1

[4] Z. Zhang, G. Wu, S. Deng, S. Wang, and Y. Zhang, "Netecho: From real-world streaming side-channels to full LLM conversation recovery," *arXiv preprint arXiv:2510.25472*, 2025, cited version: arXiv:2510.25472v1. [Online]. Available: https://arxiv.org/abs/2510.25472v1

[5] H. Jeong, M. Teymoorianfard, A. Kumar, A. Houmansadr, and E. Bagdasarian, "Network-level prompt and trait leakage in local research agents," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 6485–6504. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/jeong

[6] G. Wu, Z. Li, Y. Zhang, Z. Zhang, J. Niu, Y. Wu, and Y. Zhang, "Cacheprune: Privacy-aware and fine-grained KV cache sharing for efficient LLM inference," *arXiv preprint arXiv:2605.23640v1*, 2026. [Online]. Available: https://arxiv.org/html/2605.23640v1

[7] M. Juarez, S. Afroz, G. Acar, C. Diaz, and R. Greenstadt, "A critical evaluation of website fingerprinting attacks," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 263–274.

[8] N. Mathews, J. K. Holland, S. E. Oh, M. S. Rahman, N. Hopper, and M. Wright, "SoK: A critical evaluation of efficient website fingerprinting defenses," in *IEEE Symposium on Security and Privacy*, 2023.

[9] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Madry, and A. Kurakin, "On evaluating adversarial robustness," *arXiv preprint arXiv:1902.06705*, 2019. [Online]. Available: https://arxiv.org/abs/1902.06705v2

[10] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Advances in Neural Information Processing Systems*, vol. 33, 2020. [Online]. Available: https://arxiv.org/abs/2002.08347v2

[11] X. Cai, R. Nithyanand, T. Wang, R. Johnson, and I. Goldberg, "A systematic approach to developing and evaluating website fingerprinting defenses," in *ACM CCS*, 2014, pp. 227–238. [Online]. Available: https://doi.org/10.1145/2660267.2660362

[12] J. Gong, W. Zhang, C. Zhang, and T. Wang, "WFDefProxy: Modularly implementing and empirically evaluating website fingerprinting defenses," arXiv:2111.12629v1, 2021. [Online]. Available: https://arxiv.org/abs/2111.12629v1

[13] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, "RobustBench: a standardized adversarial robustness benchmark," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, vol. 1, 2021. [Online]. Available: https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/a3c65c2974270fd093ee8a9bf8ae7d0b-Abstract-round2.html

[14] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014. [Online]. Available: https://www.cis.upenn.edu/~aaroth/Papers/privacybook.pdf

[15] G. Smith, "On the foundations of quantitative information flow," in *Foundations of Software Science and Computational Structures*, ser. LNCS, vol. 5504, 2009, pp. 288–302. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-642-00596-1_21

[16] J. B. Almeida, M. Barbosa, G. Barthe, F. Dupressoir, and M. Emmi, "Verifying Constant-Time implementations," in *25th USENIX Security Symposium*, 2016, pp. 53–70. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/almeida

[17] E. Stefanov, M. van Dijk, E. Shi, C. Fletcher, L. Ren, X. Yu, and S. Devadas, "Path ORAM: An extremely simple oblivious RAM protocol," in *ACM SIGSAC Conference on Computer and Communications Security*, 2013, pp. 299–310. [Online]. Available: https://doi.org/10.1145/2508859.2516660

[18] J. Xu, L. Fan, Z. Wang, X. Li, and H. Liu, "Selection-invariant communication compilers for privacy-aware multi-agent LLM workflows," *arXiv preprint arXiv:2609.26076v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.26076v1

[19] J. W. Kim, B. Min, K. Edemacu, Y. Choi, S.-H. Cho, and B. Jang, "DP-MemView: A memory interface for attribute-level transcript privacy in long-term LLM agents," *arXiv preprint arXiv:2608.03130v1*, 2026. [Online]. Available: https://arxiv.org/html/2608.03130v1

[20] J. Zhu, L. Yang, R. Shan, C. Zheng, Z. Zheng, W. Liu, Y. Yu, W. Zhang, and J. Lin, "Turing test on screen: A benchmark for mobile GUI agent humanization," *arXiv preprint arXiv:2604.09574v1*, 2026. [Online]. Available: https://arxiv.org/html/2604.09574v1

[21] Kubernetes Authors, "Kubernetes API Concepts: Dry-run," https://kubernetes.io/docs/reference/using-api/api-concepts/#dry-run, accessed October 24, 2026.

[22] Stripe, "Sandboxes," https://docs.stripe.com/sandboxes, accessed October 24, 2026.

[23] PostgreSQL Global Development Group, "PostgreSQL 18 Documentation: ROLLBACK," https://www.postgresql.org/docs/18/sql-rollback.html, accessed October 24, 2026.

[24] ——, "PostgreSQL 18 Documentation: Sequence Manipulation Functions," https://www.postgresql.org/docs/18/functions-sequence.html, accessed October 24, 2026.

[25] W. Lugoloobi, S. Marro, J. Magomere, J. Wright, and C. Russell, "Known by their actions: Fingerprinting LLM browser agents via UI traces," *arXiv preprint arXiv:2605.14786*, 2026, cited version: arXiv:2605.14786v1. [Online]. Available: https://arxiv.org/abs/2605.14786v1

[26] B. Dangol, "From privacy to workflow integrity: Communication-graph metadata in autonomous agent interoperability," *arXiv preprint arXiv:2606.07150v3*, 2026. [Online]. Available: https://arxiv.org/abs/2606.07150v3

[27] N. Nazari, F. Xiang, C. Fang, H. M. Makrani, A. Puri, K. Patwari, H. Sayadi, S. Rafatirad, C.-N. Chuah, and H. Homayoun, "LLM-FIN: Large language models fingerprinting attack on edge devices," in *International Symposium on Quality Electronic Design*, 2024. [Online]. Available: https://web.cs.ucdavis.edu/~srafatir/pages/documents/llm_fingerprint.pdf

[28] H. Weng, R. Zhou, Y. Qu, X. Ji, and W. Xu, "ArchSnoop: LLM architecture snooping via electromagnetic side-channel on edge devices," in *Proceedings of the 19th ACM Conference on Security and Privacy in Wireless and Mobile Networks*, 2026, pp. 51–62. [Online]. Available: https://dl.acm.org/doi/10.1145/3765613.3811673

[29] S. Majidi, N. Mireshghallah, and K. Taram, "Leaky language models: Stealing architecture and inference optimizations via per-token timing," *arXiv preprint arXiv:2607.20723*, 2026, cited version: arXiv:2607.20723v1. [Online]. Available: https://arxiv.org/abs/2607.20723v1

[30] T. Zhang, G. Saileshwar, and D. Lie, "Time will tell: Timing side channels via output token count in large language models," *arXiv preprint arXiv:2412.15431*, 2024, cited version: arXiv:2412.15431v1. [Online]. Available: https://arxiv.org/abs/2412.15431v1

[31] H. Sun, S. Liu, S. Ma, J. Li, M. Xiao, and W. Jiang, "Agent-assisted side-channel attacks on non-prefix KV cache in rag," *arXiv preprint arXiv:2606.21842*, 2026, cited version: arXiv:2606.21842v1. [Online]. Available: https://arxiv.org/abs/2606.21842v1

[32] J. Lee, Y. Wang, R. Rajat, D. Wong, M. Li, and M. Annavaram, "Otro: Oblivious tokenization path with square-root oram," *arXiv preprint arXiv:2606.17358v1*, 2026. [Online]. Available: https://arxiv.org/html/2606.17358v1

[33] H. Ye, Y. Xia, Y. Chen, K.-Y. Chen, Y. Yuan, S. Deng, B. Kasikci, T. Mudge, and N. Talati, "Palermo: Improving the performance of oblivious memory using protocol-hardware co-design," *arXiv preprint arXiv:2411.05400v1*, 2024, cited version: v1. [Online]. Available: https://arxiv.org/html/2411.05400v1

[34] J. Xie and S. Li, "OCELOT: Inference-leakage budgets for privacy-preserving LLM agents," *arXiv preprint arXiv:2606.12341v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.12341v1

[35] G. Yu, Q. Wang, R. Lang, S. Su, and X. Wang, "PlanTwin: Privacy-preserving planning abstractions for cloud-assisted LLM agents," *arXiv preprint arXiv:2603.18377v2*, 2026. [Online]. Available: https://arxiv.org/abs/2603.18377v2

[36] P. G. Pennas, K. Papaioannou, M. Guarnieri, and T. D. Doudali, "CacheSolidarity: Preventing prefix caching side channels in multi-tenant LLM serving systems," *arXiv preprint arXiv:2603.10726v1*, 2026, cited versions: arXiv:2603.10726v1 and v2 (retitled PrefixWall). [Online]. Available: https://arxiv.org/html/2603.10726v1

[37] E. Bagdasarian, R. Yi, S. Ghalebikesabi, P. Kairouz, M. Gruteser, S. Oh, B. Balle, and D. Ramage, "AirGapAgent: Protecting privacy-conscious conversational agents," *arXiv preprint arXiv:2405.05175v2*, 2024, arXiv metadata lists ACM CCS 2024. [Online]. Available: https://arxiv.org/abs/2405.05175v2

# Appendix A. Case A Details

Defenses and attackers. The transport controls are fixed-size buckets, chunk batching, variable-stop fixed cells and a hard cap on application events; overflow-retaining minimum-count padding is a separate ablation. Observations are response application-record lengths and timing, excluding request and handshake traffic. The original attacker suite separately maximized out-of-fold BA over logistic regression, random forest and gradient boosting for each of the raw, $D_0$ and $D_2$ conditions. Selecting and reporting

each winner on the same out-of-fold predictions can be optimistic, which is why Section 5.1 reports one fixed learner instead. That learner was chosen retrospectively; its paired bootstrap (2,000 draws, 50 topics per class, all templates of each sampled topic) holds fitted predictors fixed. The five additional splits use seeds 11, 29, 47, 71 and 101 with no topic shared between training and evaluation within a split.

Calibration transfer: a negative result. We asked whether a defense parameter calibrated on one model could be imported to another. For the default bucket, batching and fixed-cell controls the question is empty: all 45 source–target–fold combinations use the same parameters (32-byte bucket quantum, four chunks per batch, 32-byte cells at 50 ms ticks), so imported-minus-local differences are zero by construction and carry no evidence. The hard-cap control does differ across sources in some folds. Table 8 reports all six imported-minus-local contrasts. Five intervals include or touch zero; the one that excludes it (Llama 1B cap imported to Qwen 8B, $+3.4$ BA points, $[0.8, 5.4]$) means worse protection, and its Holm-adjusted $p$ covers only the two imports per target. A later resampling of the 80 local training topics (500 draws per target and fold) shows how much the fitted cap itself varies: for Qwen 8B fold 0 the local cap is 87 events with a 2.5th–97.5th percentile range of 83–89, against imported caps of 90 (Qwen 0.6B) and 128 (Llama 1B). The resampling did not rerun attacks. We draw no conclusion about calibration transfer from these data beyond the design lesson in Table 2: a transfer comparison needs parameters that actually differ and a control for local fitting variability.

Run provenance. The original queue retained 34,440 successful captures, one unknown outcome and 559 jobs that never started. Recovery collected the latter and a post hoc 500-session Qwen 8B fixed-cell repeat with the same TLS implementation in a new runtime. Among 495 matched sessions, bytes and record counts agree and the median topic-mean completion difference is $-0.461$ ms; this does not rule out batch effects. The original unknown outcome remains unknown.

## Appendix B. Case B Details

An earlier run with a served GLM planner stopped after 347 of 900 planned episodes because of transport failures. Its 187 incomplete task–repeat blocks, selected by terminal-file existence rather than measured utility, were rerun in all three arms with a `qwen3.8-flash-next` planner and simulated user on independently cloned states, giving the 561 episodes over 98 tasks and 46 customers in Section 5.2. The 113 blocks that were complete at interruption (80 tasks) are reported separately and never paired with Qwen arms. Extra calls are chosen without access to gold actions, and episodes in which the cover was never triggered remain included. Task-bootstrap intervals condition on the selected cohort; sign-flip tests assume task-level sign symmetry, and the read-cover interval is not an equivalence bound. Earlier shadow checks with reference programs are separate and do not substitute for a fresh-planner side-effect-free control. Recovery changed concurrency from 10 to 2 and the transport path; recorded time excludes the ten interrupted waits and is not a complete cost.

## Appendix C. Source Material

How examples were collected. The original source map was assembled from seed papers, venue screening, citation tracing, Semantic Scholar and arXiv queries, and closed on 21 October 2026. A core source had to (C1) involve LLM or agent execution, (C2) target a hidden property, (C3) use an auxiliary observable that depends on execution, (C4) specify observer permissions and (C5) evaluate inference empirically. Because C5 requires a scored attack, the map under-represents defense papers and formal or hardware guarantees. A defense-first search on 27 October added six sources (Table 5): four named seeds (CacheSolidarity, PlanTwin, SICC, DP-MemView) and two additions (OCELOT, AirGapAgent) from nine logged queries. Hardware and formal examples (SecEmb, Palermo, OTRO, Opal, TDXRay) illustrate evidence that C5 omits; architecture venues such as ISCA, MICRO, HPCA, ASPLOS and CHES were not systematically searched. Table 9 summarizes the material. Its counts index ledgers for reconstruction; they are not samples of the literature, and the paper draws no frequency conclusions from them.

External WF check. As a sanity check, the card was applied to published WF defenses with their outcomes visible. It reproduces distinctions that the source papers already draw: WFDefProxy's FRONT comparison reports DF true-positive rate 72.06% versus 42.79% at different false-positive rates and overheads, so the observed tuples are retained rather than ranked [12, 6.2, Table 10]; DynaFlow's 379% live time overhead in Mathews et al. is not a matched simulation-versus-live effect because URLs and crawler changed [8, IV-F, Table II]. The check shows that the card is consistent with established practice, not that it finds errors that practice misses.

Version questions. Table 10 lists the version issues that affect interpretation; claims are scoped to the inspected version.

TABLE 8. ALL SIX DEFAULT HARD-CAP IMPORTED-MINUS-LOCAL PRIMARY CONTRASTS. DIFFERENCES AND 95% TOPIC-BOOTSTRAP INTERVALS ARE IN BA PERCENTAGE POINTS, CONDITIONAL ON FITTED ATTACKERS. THE SECONDARY FIXED-WINNER SIGN-FLIP $p_{\mathrm{H}}$ ADJUSTS ONLY THE TWO IMPORTS PER TARGET; IT OMITS REFITTING AND WINNER-SELECTION UNCERTAINTY. A LATER LOCAL CAP-PARAMETER RESAMPLING DOES NOT REMEASURE THE ATTACK EFFECT.

| Imported source | Target | Difference | Conditional 95% interval | $p_{\mathrm{H}}$ |
|---|---|---|---|---|
| Qwen 0.6B | Qwen 8B | $-1.2$ | $[-2.6, 1.2]$ | 0.2378 |
| Llama 1B | Qwen 8B | $+3.4$ | $[0.8, 5.4]$ | 0.0112 |
| Qwen 8B | Qwen 0.6B | $+0.6$ | $[-2.4, 1.8]$ | 0.8122 |
| Llama 1B | Qwen 0.6B | $+3.8$ | $[0.0, 6.0]$ | 0.1224 |
| Qwen 8B | Llama 1B | $+0.2$ | $[-3.0, 4.4]$ | 1.0000 |
| Qwen 0.6B | Llama 1B | $+1.0$ | $[-2.2, 4.8]$ | 1.0000 |

TABLE 9. SOURCE MATERIAL AND ITS ROLE. FULL INDICES ARE IN THE SUPPLEMENT.

| Material | Content and role |
|---|---|
| Original map | 82 source families (58 meet C1–C5, 24 adjacent); supplies most boundary examples. |
| Defense-first set | Six defense papers; Table 5. |
| Hardware/formal | Selected examples of evidence that C5 excludes. |
| External WF | 15 records from published WF defenses, used to check that the card reproduces decisions the source papers already support. |

TABLE 10. VERSION QUESTIONS RETAINED FOR SOURCE-SPECIFIC INTERPRETATION.

| Source | Status |
|---|---|
| CacheSolidarity | v1 inspected; v2 is retitled PrefixWall (latest v3). Version-specific results are not merged. |
| SICC v1 | arXiv lists submission on 4 August 2026 under a September identifier; claims scoped to v1. |
| Palermo v1 | Preprint inspected; equivalence with the published version not checked. |
| Turing Test v1 | arXiv lists submission in February 2026 under an April identifier; claims scoped to v1. |